

Chuanrui WANG

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Education

Ph.D. in Computer Science

Hangzhou, China

○ Westlake University

Starting Sep. 2026

○ Supervisor: Tailin Wu

○ Academic plans for 2024-2026 were adjusted due to visa-related changes affecting previously arranged doctoral starts at the University of Copenhagen with Ole Winther (2024) and Mila with Kirill Neklyudov (2025).

M.Sc. in Computer Science

Montréal, Canada

○ Mila / University of Montréal

Sep. 2022 - Aug. 2024

○ Supervisor: Jian Tang

○ Overall GPA: 4.3/4.0 (A+ for all courses)

○ Courses: Machine Learning, Representation Learning, Structured Optimization, AGI

○ Courses(Audit): Probabilistic Graphical Models, Reinforcement Learning and Optimal Control

Ingénieur Généraliste

Lille, France

○ École Centrale de Lille, Dual Degree Program

Aug. 2018 - Jun. 2022

○ Overall GPA: 3.70/4.0

M.Sc. in System Engineering

Beijing, China

○ Beihang University, Dual Degree Program

Aug. 2019 - Jun. 2022

○ Supervisor: Wenge Rong

○ Overall GPA: 3.64/4.0

B.Sc. in Mathematics and Applied Mathematics

Beijing, China

○ Beihang University, Entrance at age 14

Aug. 2015 - Jun. 2019

○ Major Courses GPA: 3.77/4.0, Rank: 2/120

Publications

* Equal contribution

RealPDEBench: A Benchmark for Complex Physical Systems with Real-World Data

○ Peiyan Hu, Haodong Feng, Hongyuan Liu, Tongtong Yan, Wenhao Deng, Tianrun Gao, Rong Zheng, Haoren Zheng, Chenglei Yu, Chuanrui Wang, Kaiwen Li, Zhi-Ming Ma, Dezhi Zhou, Xingcai Lu, Dixia Fan, Tailin Wu

○ In ICLR 2026 (oral).

scDFM: Distributional Flow Matching Model for Robust Single-Cell Perturbation Prediction

○ Chenglei Yu*, Chuanrui Wang*, Bangyan Liao*, Tailin Wu

○ In ICLR 2026 (poster).

GGBall: Graph Generative Model on Poincaré Ball

○ Tianci Bu*, Chuanrui Wang*, Hao Ma, Haoren Zheng, Xin Lu, Tailin Wu

○ In ICLR 2026 (poster).

PDB-Struct: A Comprehensive Benchmark for Structure-based Protein Design

○ Chuanrui Wang, Bozitao Zhong, Zuobai Zhang, Narendra Chaudhary, Sanchit Misra, Jian Tang

○ In Neurips 2023 workshops on AI for Drug Discovery/Machine Learning in Structural Biology.

A Systematic Study of Joint Representation Learning on Protein Sequences and Structures

○ Zuobai Zhang, Chuanrui Wang, Minghao Xu, Vijil C., Aurelie L., Payel Das, Jian Tang

○ In arXiv preprint.

Protein Sequence and Structure Co-Design with Equivariant Translation

○ Chence Shi, Chuanrui Wang, Jiarui Lu, Bozitao Zhong, Jian Tang

○ In ICLR 2023 (poster).

KnowReQA: A Knowledge-aware Retrieval Question Answering System

- Chuanrui Wang, Jun Bai, Xiaofeng Zhang, Cen Yan, Yuanxin Ouyang, Wenge Rong, Zhang Xiong
- In KSEM 2022.

A New Mesh Smoothing Method based on A Neural Network

- Yufei Guo, Chuanrui Wang, Zhe Ma, Xuhui Huang, Kewu Sun, Rongli Zhao
- In Comput Mech 2021 (Journal).

Colossal-AI: A Unified Deep Learning System for Large-Scale Parallel Training

- Zhengda Bian, Hongxin Liu, Boxiang Wang, Haichen Huang, Yongbin Li, Chuanrui Wang, Fan Cui, Yang You
- In arXiv preprint. ([29k+ stars on GitHub](#))

Current Interests

- My current research focuses on generative AI for science, with applications in drug discovery, cellular modeling, and scientific simulation.
- I am particularly interested in probabilistic generative models and geometric deep learning, especially diffusion models, flow matching, and transport-based methods.
- More broadly, I study how to develop structure-aware and dynamics-aware machine learning methods for complex scientific systems.

Research Experiences

Research in Generative Models and Periodic Crystal Generation

Advisor: Tailin Wu, Assistant Professor, Westlake University

Oct. 2024 - Present

- Collaborating with Kirill Neklyudov, Assistant Professor, Mila.
- Developing a flow matching based generative model for periodic crystal structures.

Research in Generative Models and Protein Design

Advisor: Jian Tang, Associate Professor, Mila

Jun. 2022 - Present

- Developed a novel protein sequence-structure co-design method with rapid one-shot inference, overcoming the slow inference issues of autoregressive and diffusion generative models. We demonstrated its high sampling quality and efficiency, alongside flexible conditioning in tasks such as antibody CDR design and general protein design. The work was accepted at ICLR 2023, where I also presented a poster session.
- Improved the evaluation of protein sequence design models by creating two innovative metrics: refoldability, which uses structure prediction models as proxies for wet-lab experiments, and stability, to evaluate the likelihood landscapes of generated proteins. Furthermore, we introduced the PDB-Struct benchmark with carefully selected datasets to facilitate the comparison of recent protein design methods, thereby aiding protein scientists in model selection. This work was accepted by NeurIPS 2023 workshops focusing on AI for Drug Discovery and for Structural Biology.
- In progress: Designing a diffusion-based generative model that simultaneously designs sequences and structures, which differs from previous models that operate in a sequential manner. The challenge lies in designing a diffusion process that operates within combination of discrete space and continuous SE(3) space.

Research in Protein Representation Learning and Protein Folding

Advisor: Jian Tang, Associate Professor, Mila

Sep. 2022 - May. 2023

- Investigated protein representation learning by integrating sequence and structural data through a state-of-the-art protein language model and various structural encoders.
- Proposed three fusion strategies and six pre-training objectives for constructing and training such joint learning model. The enhanced model significantly outperformed sequence-based and structure-based models in function prediction tasks. These findings are detailed in a paper under review for AAAI 2024.
- Improved protein folding models using pseudo-data backtranslated from an inverse protein folding model, aiming to overcome the scarcity of labeled sequence-structure pair data. Experiments yielded enhanced results with antibody datasets; however, we have yet to replicate this success with general protein datasets, possibly constrained by the availability of computational resources.

Large-Scale Model Training System with Efficient Parallelization Techniques

Advisor: Yang You, Professor, National University of Singapore

Aug. 2021 - Feb. 2022

- Developed [Colossal-AI](#), a distributed machine learning framework. The framework enabled hybrid parallel training with minimum pytorch code modification, enhancing the scalability of AI model training.
- Realized the 1d tensor parallel architecture, conducted scaling experiments, and achieved efficient pre-training of a ViT-base/32 model in 0.6 hours, showcasing the system's effectiveness.

Research in Natural Language Processing and Pre-trained Language Models

Advisor: Wenge Rong, Professor, Beihang University

Nov. 2020 - Aug. 2021

- Proposed a Hierarchical Lexicon Embedding Structure to better solve Chinese named entity recognition (NER) task and outperformed baselines in three commonly used Chinese NER dataset (MSRA, Resume and Weibo).
- Proposed a Question Answer System that fuses Knowledge Graph information using an attention-based filter mechanism, improving detail and context in natural language responses.

Skills

- Proficient Skills: Python, Pytorch, Linux, Latex, Slurm
- Intermediate Skills: C, SQL, HTML, Matlab, Office
- Language Proficiency: Mandarin (Native); English (IELTS 8.0); French (DELF B2)
- Academic Service: Reviewer for NeurIPS, ICLR, and AISTATS (since 2024)